

# Moloco: Bringing ML- powered retail media advertising technology to the world with rapid global scaling

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ABOUT MOLOCO



## About Moloco

By building its ML advertising service on Google Cloud, Moloco has found the infrastructure systems it needs to provide its customers a rapid, highly scalable platform.

Moloco's mission is to empower businesses of all sizes to grow through operational machine learning. With Moloco's machine learning platform for growth and performance, every app publisher and e-commerce marketplace can now unlock the value of their unique, first-party data. Moloco was founded in 2013 and has offices throughout the US, the UK, Germany, Korea, China, India, Japan, and Singapore.

Industries: Retail & Consumer Goods, Technology

## Google Cloud results

- Supporting more than 35,000 advertisers with Moloco RMP
- Millisecond speed delivery from model to ad unit
- 20+ deep neural network models drive Moloco's ad tech services

Moloco processes around 1 billion requests a day, catering to a vast, competitive world, supporting real-time, data-driven decisions.

Location: Korea, the US, the UK, China, Japan, India, and Singapore

(<https://cloud.google.com/armor/>), [Cloud Bigtable](https://cloud.google.com/bigtable/) (<https://cloud.google.com/bigtable/>), Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

(<https://cloud.google.com/scheduler/>), [Cloud](https://cloud.google.com/armor/)

Google Cloud Armor

(<https://cloud.google.com/armor/>)

Kubernetes Engine

Cloud Bigtable (<https://cloud.google.com/bigtable/>)

Cloud SQL (<https://cloud.google.com/sql/>)

(<https://cloud.google.com/pubsub/>),

Cloud Scheduler

(<https://cloud.google.com/scheduler/>)

Enterprise

Cloud Spanner (<https://cloud.google.com/spanner/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

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Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

TensorFlow Enterprise

(<https://cloud.google.com/tensorflow-enterprise>)

The retail landscape has never been more competitive, both for stores and their suppliers. Profitability takes careful investment in the right technologies and marketing strategies. For machine learning company [Moloco](https://www.moloco.com/) (<https://www.moloco.com/>), finding a way to deliver access to such technologies led to becoming one of the fastest-growing ad tech companies in history.

Founded in 2013 and built on Google Cloud since inception, Moloco's mission is to empower businesses of all sizes to grow through operational machine learning. It has three core services today. Moloco Cloud DSP enables performance marketers to quickly scale user acquisition and achieve greater lifetime value through market validated prediction models. Moloco Retail Media Platform (RMP) enables commerce platforms to establish their own performance ads business. Moloco's monetization solution for streaming media and OTT providers helps content owners increase revenue by building unique ad businesses enabling them to serve a broader set of customers than in traditional linear TV businesses.

"It's not only  
about making  
excellent  
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It's also about  
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surrounding  
systems. By  
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*—Bill Michels, GM of RMP,  
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With its RMP offering, Moloco leverages its operational machine learning to empower marketplaces and retailers to maximize ad revenue. It does this through automating and scaling ads onboarding, buying, and operations so that it can deliver relevant user ads and optimize advertiser performance. Today, it enables and delivers performance to more than 35,000 advertisers globally.

For Bill Michels, Moloco General Manager of Retail Media Platform, a critical factor in Moloco's success was in knowing they needed to choose the right infrastructure as the bedrock of their platform.

"It's not only about building and training excellent machine learning models," says Michels. "It's also about developing surrounding systems. We had to build scalable online ad-serving infrastructure, reliable real-time user behavioral log ingestion pipelines, continuous training / continuous deployment ML training pipelines, a secure advertising campaign manager, a real-time advertising bidding engine, and more. By choosing

Google Cloud and its suite of services suited to each purpose, we delivered the RMP on time."

## **Data performance at scale in minutes**

Considering the company uses a wide range of Google Cloud solutions, Moloco considers BigQuery (<https://cloud.google.com/bigquery>) to be the standout capability that gives its RMP the most advantages.

"We needed data gathering and analytics tools to operate the high-performance ML-based service," says Michels. "Using BigQuery, we built reliable data pre-processing pipelines to feed data into ML training pipelines and allow data scientists and analysts to run data experiments and generate ad performance reports for the teams and customers."

Moloco has found that the elastic infrastructure capacity and geographic distribution of Google Cloud gives the company the ability to rollout into new global regions within weeks instead of months. It makes it simple for the company to be available wherever a customer wants them to be while always delivering on its promise of highly targeted ML performance.

"We use Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE) and the autoscaler to increase and decrease the infrastructure capacity in minutes," says Michels. "Cloud TPU (<https://cloud.google.com/tpu>) trains the transformer-based deep learning models to build the

hyper-personalized recommendation service in a continuous training/deployment fashion."

Moloco has found the Google Cloud team to be very supportive of the company's efforts to get its RMP working in highly specific ways, stepping in to assist in an "above and beyond" capacity to help.

"When we needed a reliable way to mirror the traffic coming into our LeaderBoards (LBs), Google Cloud was willing to support a new feature earlier than originally scheduled," says Michels. "While the task itself was part of the roadmap, we were impressed that they were able to prioritize our needs and change their schedule based on our feedback."

## **Millisecond performance with ML data in Cloud Bigtable**

Dynamic digital retail is at its best when algorithmic choices are made with near instant speeds.

Ensuring each customer sees the right recommendation ensures each advertiser has the highest opportunity for sales and the marketplace sees the best effectiveness per visit.

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At Moloco, its EMP gathers real-time behavioral logs for each shopper based on details like keyword searches, product detail page visits, and purchased products. To use this data in its ML training to deliver real-time product recommendations most effectively, Moloco uses Cloud Bigtable (<https://cloud.google.com/bigtable>).

"We store all our behavioral data in Cloud Bigtable for retrieval by RMP. The low latency and scalability of Cloud Bigtable enabled us to serve sponsored advertising recommendations within milliseconds," says Michels.

Like all data-driven services, the proof is often clear in the numbers. Moloco has integrated its RMP with a leading South Korean retailer, Bucketplace

(<https://www.bucketplace.com/en/>), which prides itself on excellence in user experience. Bucketplace recently published that "AI advertisement program, which first started in April of last year, recorded more than 3 billion exposures to the company's products for a year, and conversion sales through advertising exceeded 100 billion won. Small and medium-sized businesses that used the advertising program showed an average increase of 2.1 times in today's house in 2022 compared to 2021." [[source](https://www.techm.kr/news/articleView.html?idxno=111202) (<https://www.techm.kr/news/articleView.html?idxno=111202>) ]

## **Managing data sovereignty requirements with cloud regions**

While Moloco has embraced the speed of scaling through Google Cloud's global infrastructure, it is also acutely aware that customer data must be carefully managed to adhere to data sovereignty requirements. Depending on where a marketplace's customers reside, rules must be stipulated for how their data is managed and where it can be moved or accessed.

"We appreciate how Google Cloud helps us quickly meet these data sovereignty needs in various new countries," says Michels. "For example, we had an EU customer that required data transfers to be confined to certain regions and Google Cloud allowed us to quickly satisfy these needs. Also, since Google launched the new region in Israel in October 2022, we are now able to actively evaluate



customer needs in Israel and the Middle East Asia markets."

## **Small teams, big results**

"Moloco RMP would not exist without the support of Google Cloud," says Michels. "RMP did not start with a lot of engineers, and because of the ease of use and abstraction layers provided by Google Cloud, we were able to start a complex service in a short period of time."

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"Like any product build, there were bumps along the way. But Google Cloud was there to support us when issues arose and we were able to ultimately build a better product thanks to their tools and support."

Looking ahead, Moloco expects to continue its global expansion of availability, as well as to help its customers also scale their businesses and to help them seek greater profitability by launching a new profit center tool to further assist its customers to maximize the potential of RMP advertising in their businesses.